



Inventory Management



Introduction

What is Inventory??

Inventory is stock or store of goods



3 types of Inventory

- Raw material
- Work-in-process
- Finished goods

MROs (Maintenance, Repair, and Operating Materials)

Independent Versus Dependent Demand

- **Independent demand** - the demand for item is independent of the demand for any other item in inventory
- **Dependent demand** - the demand for item is dependent upon the demand for some other item in the inventory



Dependent demand



Independent demand



Functions of Inventory

- To meet anticipated demand
- To smooth production requirements - seasonal
- To protect against stock outs
- To take advantages of quantity discounts



Objectives of Inventory Control

Inadequate control of inventories can result in both under and overstocking of items

The overall objective of inventory management is to achieve **satisfactory levels of customer service** while keeping **inventory costs within reasonable bounds**.



The *timing*



Size of orders



Inventory Costs



- **Holding (carrying) cost**

Cost to carry an item in inventory for a length of time.

- **Ordering costs**

Costs of ordering and receiving inventory.

- **Setup costs**

The costs involved in preparing equipment for a job.

- **Shortage costs**

Costs resulting when demand exceeds the supply of inventory; often unrealized profit per unit.

****Lead time:** Time interval between ordering and receiving the order.



Classification System

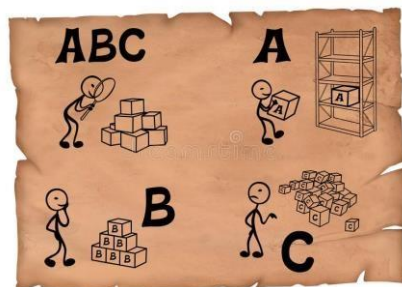
ABC Analysis

- Classifies inventory items according to some measure of importance and then allocates control efforts accordingly.
- The origin of ABC analysis is “*The Pareto Principle*” which is named after an Italian economist Vilfredo Pareto, also called as 80/20 rule. This principle suggests that 80% of the total output is generated only by 20% of the valuable efforts.

ABC Analysis

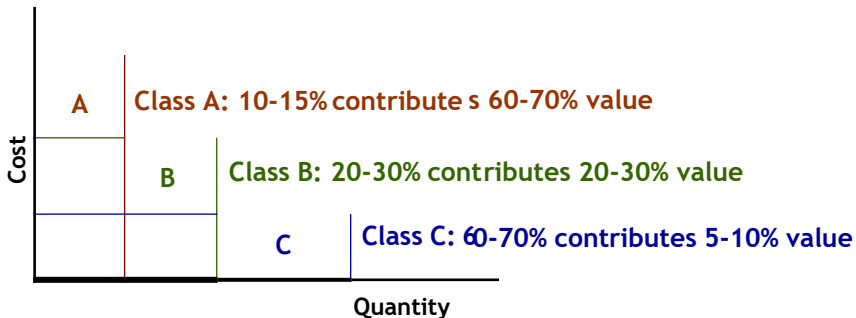
Typically, three classes of items:

- A (very important)
- B (moderately important)
- C (least important)





ABC Analysis





ABC Analysis Example-1

A manager has obtained a list of unit costs and estimated annual demands for 10 inventory items and now wants to categorize the items on an A-B-C basis.

Items	Annual Demand	Cost per unit (£)
Boxes of Paperclip	21,000	0.50
Boxes of staples	10,000	0.50
Correcting fluid	16,000	1.50
Diaries	50,000	3.50
Erasers	15,000	0.10
Notepads	40,000	2.0
Pencils	80,000	0.03
Pens	1,20,000	0.05
Rulers	15,000	0.25
Staplers	10,000	1.50



ABC Analysis Example-1

Item	Annual number of units sold	Cost per unit	Annual consumption value
Boxes of paperclips	21,000	£0.50	£10,500
Boxes of staples	10,000	£0.50	£5,000
Correcting fluid	16,000	£1.50	£24,000
Diaries	50,000	£3.50	£175,000
Erasers	15,000	£0.10	£1,500
Notepads	40,000	£2.00	£80,000
Pencils	80,000	£0.03	£2,400
Pens	120,000	£0.05	£6,000
Rulers	15,000	£0.25	£3,750
Staplers	10,000	£1.50	£15,000



ABC Analysis Example-1

Item	Annual number of units sold	Cost per unit	Annual consumption value
Diaries	50,000	£3.50	£175,000
Notepads	40,000	£2.00	£80,000
Correcting fluid	16,000	£1.50	£24,000
Staplers	10,000	£1.50	£15,000
Boxes of paperclips	21,000	£0.50	£10,500
Pens	120,000	£0.05	£6,000
Boxes of staples	10,000	£0.50	£5,000
Rulers	15,000	£0.25	£3,750
Pencils	80,000	£0.03	£2,400
Erasers	15,000	£0.10	£1,500



ABC Analysis Example-1

Item	Annual number of units sold	Cost per unit	Annual consumption value	% Of annual units sold	% Of annual consumption value
Diaries	50,000	£3.50	£175,000	13.3%	54.2%
Notepads	40,000	£2.00	£80,000	10.6%	24.8%
Correcting fluid	16,000	£1.50	£24,000	4.2%	7.4%
Staplers	10,000	£1.50	£15,000	2.7%	4.6%
Boxes of paperclips	21,000	£0.50	£10,500	5.6%	3.2%
Pens	120,000	£0.05	£6,000	31.8%	1.9%
Boxes of staples	10,000	£0.50	£5,000	2.7%	1.5%
Rulers	15,000	£0.25	£3,750	4.0%	1.2%
Pencils	80,000	£0.03	£2,400	21.2%	0.7%
Erasers	15,000	£0.10	£1,500	4.0%	0.5%
Totals	377,000		£323,150		

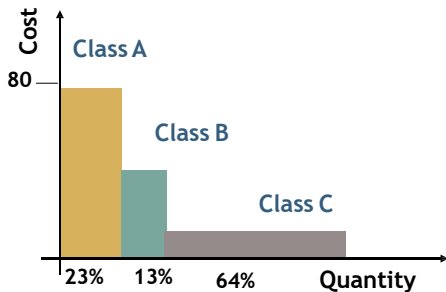


ABC Analysis Example-1

	Item	Annual number of units sold	Cost per unit	Annual consumption value	% of annual units sold	% of annual consumption value
79%	Diaries	50,000	£3.50	£175,000	13.3%	54.2%
	Notepads	40,000	£2.00	£80,000	10.6%	24.8%
15%	Correcting fluid	16,000	£1.50	£24,000	4.2%	7.4%
	Staplers	10,000	£1.50	£15,000	2.7%	4.6%
	Boxes of paperclips	21,000	£0.50	£10,500	5.6%	3.2%
6%	Pens	120,000	£0.05	£6,000	31.8%	1.9%
	Boxes of staples	10,000	£0.50	£5,000	2.7%	1.5%
	Rulers	15,000	£0.25	£3,750	4.0%	1.2%
	Pencils	80,000	£0.03	£2,400	21.2%	0.7%
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ABC Analysis Example-1





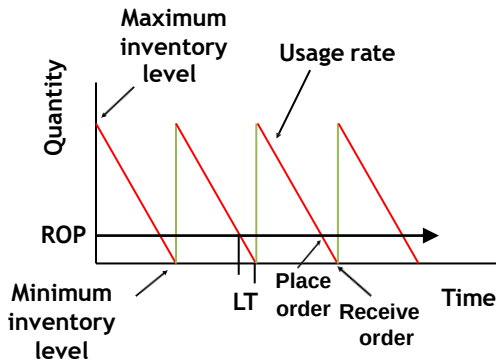
Inventory Models for Independent Demand

Need to determine when and how much to order

- The basic economic order quantity model
- The economic production quantity model
- The quantity discount model



Inventory Ordering & Usage Cycle



Economic Order Quantity (EOQ)

EOQ is the order quantity that minimizes total cost

Assumptions

- Only one product is involved
- Annual demand requirements are known
- Demand is spread evenly
- Lead time does not vary
- No quantity discount



Cost Calculations

$$\text{Annual Holding Cost, } HC = \frac{Q}{2} H$$

Average inventory level $\times$ Holding cost per unit per year

$$\text{Annual Ordering Cost, } OC = \frac{D}{Q} S$$

Number of orders placed annually $\times$ Order cost per order

$$\text{Total Cost, } TC = \frac{Q}{2} H + \frac{D}{Q} S$$

$$\text{Total Cost (including Material), } TC = \frac{Q}{2} H + \frac{D}{Q} S + \boxed{D \times P}$$

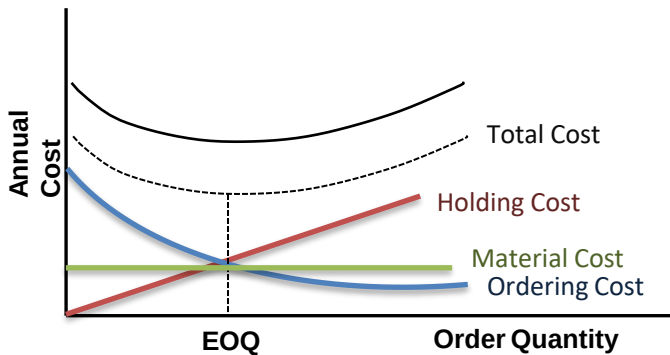
Material Cost

$$\text{for Minimum Cost, } Q = \sqrt{\frac{2DS}{H}}$$





Cost Curves





EOQ Example – 1

(LT = 0)

A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.

- Determine the optimal number of tires to order.



EOQ Example – 1

(LT = 0)

A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.

a. Determine the optimal number of tires to order.

$$\text{We know, } Q = \sqrt{\frac{2D}{H}} = \sqrt{\frac{2 \times 9600 \times 75}{16}} = 300 \text{ tires per order}$$



EOQ Example – 1

*A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.*

b. How many times per year does the store reorder?



EOQ Example – 1

A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.

b. How many times per year does the store reorder?

We know,

$$\begin{aligned}\text{Number of orders per year, } N &= \frac{D}{Q} = \frac{9600 \text{ tires/yr}}{300 \text{ tires/order}} \\ &= \mathbf{32 \text{ orders per year}}\end{aligned}$$



EOQ Example – 1

*A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.*

- c. Determine the length of an order cycle or expected time between orders.



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A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.

c. Determine the length of an order cycle or expected time between orders.

$$\begin{aligned}\text{Length of an order cycle} &= \frac{\text{Number of working days per year}}{\text{Number of orders per year, } N} \\ &= \frac{288}{32} = 9 \text{ days}\end{aligned}$$



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A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.

d. Determine the annual Total cost.



EOQ Example – 1

A local distributor for a national tire company expects to sell approximately **9,600 steel-belted** radial tires of a certain size and tread design next year. Annual carrying cost is **\$16 per tire**, and ordering cost is **\$75**. The distributor operates **288 days a year**.

d. Determine the annual Total cost.

We know,

$$\begin{aligned} \text{Total Cost, } TC &= \frac{Q}{2} H + \frac{D}{Q} \\ &= \left(\frac{300}{2} \times 16 \right) + \left(\frac{9600}{300} \times 75 \right) \\ &= 2400 + 2400 = \mathbf{\$4800} \end{aligned}$$



EOQ Example – 2

(LT = 0)

*Piddling Manufacturing assembles security monitors. It purchases **3,600 black-and-white cathode ray tubes** a year at **\$65 each**. Ordering costs are **\$31**, and annual carrying costs are **20 percent of the purchase price**.*

- Compute the optimal quantity and the total annual cost of ordering and carrying the inventory.



EOQ Example – 2

(LT = 0)

*Piddling Manufacturing assembles security monitors. It purchases **3,600 black-and-white cathode ray tubes** a year at **\$65 each**. Ordering costs are **\$31**, and annual carrying costs are **20 percent of the purchase price**.*

- a. Compute the optimal quantity and the total annual cost of ordering and carrying the inventory.

$$\begin{aligned} \text{We know, } Q &= \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \times 3600 \times 31}{65 \times 0.2}} = \sqrt{\frac{2 \times 3600 \times 31}{13}} \\ &\approx 131 \text{ cathode ray tubes} \end{aligned}$$



EOQ Example – 2

*Piddling Manufacturing assembles security monitors. It purchases **3,600 black-and-white cathode ray tubes** a year at **\$65 each**. Ordering costs are **\$31**, and annual carrying costs are **20 percent of the purchase price**.*

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EOQ Example – 2

*Piddling Manufacturing assembles security monitors. It purchases **3,600 black-and-white cathode ray tubes** a year at **\$65 each**. Ordering costs are **\$31**, and annual carrying costs are **20 percent of the purchase price**.*

- a. Compute the optimal quantity and the total annual cost of ordering and carrying the inventory.

We know,

$$\begin{aligned} \text{Total Cost, } TC &= \frac{Q}{2}H + \frac{D}{Q}S = \left(\frac{131}{2} \times 13\right) + \left(\frac{3600}{131} \times 31\right) \\ &\approx 852 + 852 = \mathbf{\$1704} \end{aligned}$$



A large bakery buys flour in **25-pound bags**. The bakery uses an average of **1,215 bags** a year. Preparing an order and receiving a shipment of flour involves a cost of **\$10 per order**. Annual carrying costs are **\$75 per bag**.

- Determine the economic order quantity.
- How many orders per year will there be?
- Compute the total cost of ordering and carrying flour.
- If holding costs were to increase by \$9 per year, how much would that affect the minimum total annual cost?



Solution



a. Economic Order Quantity:

$$\text{We know, } Q = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \times 1215 \times 10}{75}} = \mathbf{18 \text{ bags per order}}$$



Solution



b. Number of orders per year:

$$\frac{D}{Q} = \frac{1215}{18} = 67.5$$

c. Total cost:

$$\begin{aligned} TC &= \frac{18}{2}H + \frac{1215}{18}S = \left(\frac{18}{2} \times 75\right) + \left(\frac{1215}{18} \times 10\right) \\ &= 675 + 675 = \mathbf{\$1350} \end{aligned}$$

Solution



d. By increasing holding costs by \$9 per year:

$$Q = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \times 1215 \times 10}{75 + 9}} = \sqrt{\frac{24300}{84}} = 17 \text{ bags per order}$$

$$\begin{aligned} \therefore \text{Total Cost, } TC &= \frac{Q}{2}H + \frac{D}{Q}S = \left(\frac{17}{2} \times 84\right) + \left(\frac{1215}{17} \times 10\right) \\ &= 714 + 715 = \mathbf{\$1429} \end{aligned}$$

Thus, total annual cost increases by **\$79**



Reorder Point (ROP)

If demand and lead time are both constant, the reorder point is easy to calculate.

$$ROP = d \times LT$$

Where,

d = Demand rate (units per day or week)

LT = Lead time in days or weeks



Reorder Point (ROP) Example – 1

(LT $\neq$ 0)

Kevin takes **Two-a-Day vitamins**, which are delivered to his home by a routeman **three days** after an order is called in. At what point should Kevin reorder?

Usage rate = **2 vitamins per day**

Lead time = **3 days**

ROP = Usage rate $\times$ Lead time
= 2 vitamins per day $\times$ 3 days
= **6 vitamins**



Safety Stock

Stock that is held in excess of expected demand due to variable demand and/or lead time.

$$ROP = d \times LT + ss$$

Where,

d = Demand rate (units per day or week)

LT = Lead time in days or weeks

ss = Safety Stock



ABC Analysis Example

Part Number	Description	Annual Demand	Unit Value (\$)
1	Boxes	500	3
2	Cardboard	18000	0.02
3	Cover Stock	10000	0.75
4	Glue	75	40
5	Inside Cover	20000	0.05
6	Reinforcing Tape	3000	0.15
7	Ink	150000	0.45



*Thank
You*

Reference: Operations Management (11th edition) By William Stevenson (**Chapter 13**)